higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

ELECTRICAL SYSTEMS AND CONSTRUCTION
NQF LEVEL 3

(12041033)

21 November 2019 (X-Paper)
09:00–12:00

This question paper consists of 9 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each question on a NEW page.
5. Use a pen with only BLUE or BLACK ink.
6. Write neatly and legibly.

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QUESTION 1

1.1 Define the term *certificate of compliance* (CoC). (3)

1.2 People, animals and property must be protected against harmful earth fault currents by implementing protective measures such as electrical separation of circuits.

Explain what is meant by *electrical separation of circuits.* (2)

1.3 Study the diagram (FIGURE 1) below:

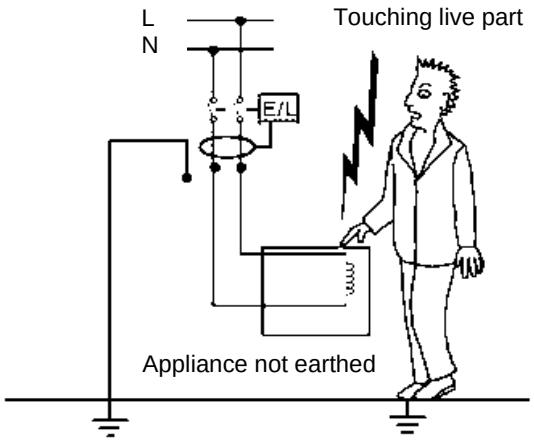


FIGURE 1

1.3.1 Explain what is meant by *indirect contact shock*.

1.3.2 Give TWO ways in which electric shock by indirect contact can be prevented.

(2 × 2) (4)

1.4 Draw a neat, labelled diagram of a single-phase stove circuit with an isolator. (5)

1.5 Give the recommended wire size and circuit breaker rating for a lighting circuit. (2)

1.6 Explain the purpose of the following metal conduit accessories:

1.6.1 Coupling

1.6.2 Male bush

(2 × 1) (2)

1.7 Name the tool required for installing conductors into a conduit. (1)

1.8 When installing cables, safety and cost should be the first consideration for any electrician.

State THREE guidelines to follow when installing electrical cables. (3 × 2) (6)

[25]

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QUESTION 2

2.1 State TWO benefits of installing prepaid meters. (4)

2.2 A distribution board is an enclosure that contains electrical equipment for the distribution or control of electrical power from one or more incoming circuits to one or more outgoing circuits.

Name FOUR components that can be found in a residential distribution board. (4)

2.3 After an installation has been wired, it has to be inspected and tested.

State SIX inspections that need to be carried out on an electrical installation by a registered person. (6)

2.4 Give TWO common causes of taking incorrect measurements when testing electrical installations. (2)

2.5 An insulation resistance test is done to ensure that the insulation of conductors is in a good condition and will prevent excessive leakage currents.

Draw a neat, labelled diagram to show how an insulation resistance test can be done of the conductors in a distribution board. (5)

2.6 Explain what must be done if the elevated voltage between incoming neutral and external earth exceeds 50 V. (2)

2.7 Calculate the maximum voltage drop allowed at a socket outlet if the supply voltage is 230 V. (2)

[25]

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QUESTION 3

3.1 After an automatic electric kettle has been switched on, the water does not heat up.

Give TWO possible causes for the water not heating up. (2)

3.2 Give TWO types of injuries that may occur when maintaining lighting systems. (2)

3.3 List FOUR symptoms that may indicate the presence of a problem on an electrical appliance. (4)

3.4 State a possible cause of and a corrective action to be taken for the following faults:

3.4.1 The voltage measured for the stove is 216, 5 V under load.

3.4.2 230 V is measured at the geyser element, but the water is not heating up.

(2 × 2) (4)

3.5 Safe operating procedures are written guidelines for procedures and tasks involving recognised hazards.

State THREE types of tasks for which safe operating procedures can be written. (3)
[15]

QUESTION 4

Various options are given as possible answers to the following questions. Choose the correct answer and write only the letter (A–D) next to the question number (4.1–4.10) in the ANSWER BOOK.

4.1 In South Africa, electricity is normally distributed by means of a ...

A 230-V/400-V, three-phase, four-wire system with a tolerance of approximately 10% of these voltages.
B two-wire DC supply of 220 V.
C 525-V, three-phase, three-wire system with a tolerance of approximately 10% of this voltage.
D 1 000-V, three-phase, four-wire system.

4.2 Low voltage (LV) refers to nominal voltage levels up to and including …

A 6,6 kV.
B 1 kV.
C 11 kV.
D 33 kV.

4.3 The TWO types of distribution network connections are …

A primary and secondary distribution systems.
B underground and overhead LV networks.
C radial and ring distribution systems.
D series and parallel feeder networks.

4.4 ... is a statement that confirms that a circuit or item of equipment is safe to work on and specifies exactly what work must be carried out.

A An isolation procedure
B A maintenance procedure
C Safe practice
D A work permit

4.5 The main purpose of earthing is …

A to ensure that no person operating an electrical installation can receive an electric shock that could cause an injury or fatality.
B to maintain the potential of any part of a system at a definite value with respect to earth.
C to allow a current to flow to earth in the event of a fault to ensure that protective devices will operate to isolate the faulty circuit.
D to ensure that an apparatus cannot reach a dangerous potential with respect to earth.

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4.6 Which of the following materials is most suitable for earth electrodes?

A Bare aluminium
B Bare copper
C Mica
D Mild steel

4.7 The letter 'S' in the TN-S earthing systems means …

A one or more parts are connected directly to earth.
B all live parts are isolated from earth, or one point is connected to earth through an impedance.
C the neutral and protective functions on the LV distributor and in the consumer's electrical installation are provided by separate conductors.
D the neutral and protective functions on the LV distributor and in the consumer's electrical installation are combined in a single conductor.

4.8 The first step in isolating equipment and machinery is to ...

A review the permit to work.
B put up notices such as a tag out.
C test the installation.
D think, check and plan.

4.9 ... are used to attach voltage detectors and short-circuiting equipment to overhead lines safely.

A Operating sticks
B Insulating gloves
C Fuse links
D Low-voltage insulating mats

4.10 Which of the following is not part of restoring the supply to an LV network?

A The completed work is tested and checked against the work schedule.
B The supply is isolated and locked out.
C Tools, equipment and any surplus resources are cleaned, checked and returned to storage.
D All workers are informed that the apparatus will be returned to service according to the regulations.

(10 × 1) **[10]**

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QUESTION 5

5.1 Name the material that is used in the construction of incandescent lamp filaments. (1)

5.2 State TWO types of filaments used in incandescent lamps. (2)

5.3 Draw a neat, labelled circuit diagram for a high-pressure sodium vapour lamp. (**NOTE:** No internal wiring of the lamp is required.) (5)

5.4 Study the diagram in FIGURE 2 and label the different parts marked 5.4.1–5.4.3. Write only the answer next to the question number (5.4.1–5.4.3) in the ANSWER BOOK.

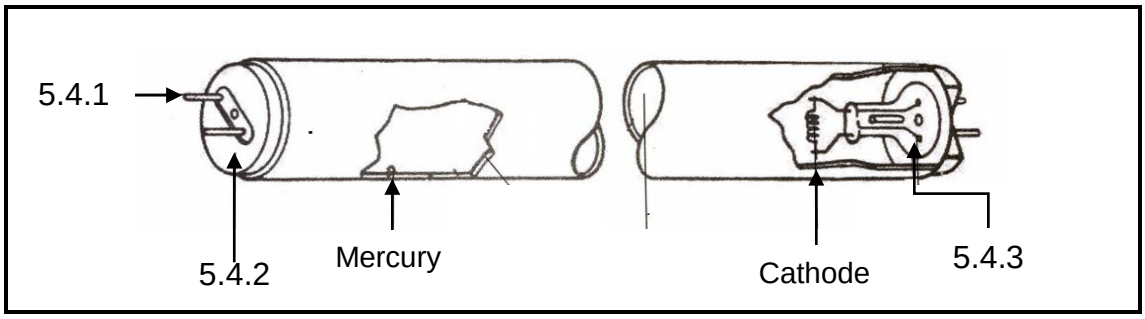


FIGURE 2 (3)

5.5 Give TWO applications in which fluorescent lamps are used. (2)

5.6 State TWO disadvantages of a low-power factor. (2)

[15]

QUESTION 6

6.1 Ladders should be subjected to a detailed visual inspection in order to identify defects that could cause serious injuries.

Give TWO defects that usually occur on step ladders. (2)

6.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (6.2.1–6.2.8) in the ANSWER BOOK.

6.2.1 The total irradiance striking the earth on a clear day is about 2 kW/m^2.

6.2.2 The physical process in which a solar cell converts sunlight into electricity is called the photovoltaic effect.

6.2.3 The nature of a photovoltaic module depends on the cell material, glazing material and electrical connections.

6.2.4 A typical silicon solar cell produces 2,5 V.

6.2.5 A blocking diode is used to prevent a battery from discharging when there is no sunlight.

6.2.6 The total voltage produced by three 0,5-V cells connected in series is 1,5 V.

6.2.7 The function of a photovoltaic inverter is to prevent batteries from overcharging.

6.2.8 One of the disadvantages of solar panels is that they require regular maintenance, which is very costly.

(8 × 1) (8)

[10]

TOTAL: 100

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